Four-Corner Spectral Tuning for Search-Free ADMM in Diffeomorphic Image Registration

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August 2026

Abstract

ADMM parameter choice strongly affects the runtime of variational image registration. For a well-posed quadratic subproblem, every admissible ADMM penalty and relaxation is already contractive; the practical task is therefore to select a rapidly convergent pair without numerical spectral search. The rank-one pixelwise registration Hessian and scalar Sobolev symbol reduce the constant-model tuning problem exactly to four curvature–symbol corners. A finite algebraic candidate set consequently yields one-shot joint tuning of (ρ, α). On 134 FIRE retinal pairs at 256×256, the predictor reduced median pairwise runtime by 34.4% (95% bootstrap CI 30.7–35.7%) and inner iterations by 39.6% relative to an externally validation-selected fixed pair; it was also 51.7% faster than a safeguarded BB proxy. Landmark error and discrete-Jacobian statistics were unchanged. On ten CIMA histology pairs, matched-protocol runtime decreased by 20.1%. Signed variable-coefficient enclosures provide optional rigorous rate bounds, but are conservative and are not part of the timed algorithm.

1 Introduction

Diffeomorphic image registration estimates an invertible, topology-preserving spatial transformation that aligns a moving image with a fixed image. Classical methods remain attractive because they provide explicit control over the deformation model, regularization, and numerical stopping criteria, and because they can be applied without training data. Representative approaches include DARTEL, diffeomorphic demons, large-deformation optimal-control formulations, and modern GPU-accelerated variational solvers [1, 15, 10, 14]. Their main practical limitation is computational cost.

Many diffeomorphic schemes update a velocity field through a sequence of convex, linearized optical-flow problems. A typical quadratic subproblem is

$$\min_{v=(v_x,v_y)} \frac{\mu}{2} \left\| I_x \odot v_x + I_y \odot v_y + I_t \right\|_2^2 + \frac{1}{2} \left\| Lv \right\|_2^2, \tag{1}$$

where I_x, I_y are spatial image derivatives, I_t is the current temporal or residual derivative, and L is a differential regularization operator. Splitting the data and regularization terms produces inexpensive pixelwise and Fourier-domain subproblems, making ADMM particularly suitable for GPU implementation [14, 12].

The penalty ρ and relaxation α strongly affect convergence. Fixed defaults, residual balancing, and adaptive spectral rules can be effective, but they do not exploit the unusually specific geometry of registration: the data Hessian is block diagonal with rank-one 2×2 blocks, while the regularizer is a scalar elliptic operator applied identically to each velocity component. General quadratic ADMM theory derives optimal parameters for special classes [5, 13, 6]; more recently, Song et al. optimized

ρ numerically for general linear quadratic problems and derived an optimal α conditional on ρ, including experiments on diffeomorphic registration [12]. Their registration penalty still requires spectral optimization.

Our answer is based on local Fourier analysis (LFA), a classical tool for predicting and optimizing stationary iterations for constant-coefficient differential operators [4, 16]. Standard LFA is exact for translation-invariant periodic problems, while nonstandard variants can remain predictive for jumping and random coefficients [9, 11]. Here, the coefficients that vary are the image-gradient blocks in the data Hessian. We freeze those blocks over patches, derive the exact patchwise ADMM symbol, and then control the discrepancy to the full iteration by a resolvent perturbation argument.

Contributions.

1. An exact reduced Cayley representation and strict-contraction characterization for well-posed quadratic registration subproblems.
2. An exact four-corner reduction and finite algebraic, search-free joint predictor for rank-one registration Hessians.
3. Optional signed variable-coefficient rate enclosures for nonnormal iterations.
4. FIRE and CIMA validation demonstrating lower end-to-end runtime at matched landmark accuracy and topology.

2 Related Work

2.1 ADMM parameter selection

The scaled ADMM iteration and practical residual stopping criteria were systematized by Boyd et al. [2]. Ghadimi et al. derived optimal penalties and relaxation parameters for several quadratic classes [5]. Teixeira et al. studied scaling, preconditioning, and optimal ADMM parameters in distributed quadratic programming [13]. Giselsson and Boyd developed tight contraction bounds and metric selection for Douglas–Rachford splitting and ADMM under strong-convexity and smoothness assumptions [6]. Adaptive alternatives include residual balancing and Barzilai–Borwein-type spectral updates [17, 18].

Song et al. [12] is the closest work. It formulates ADMM/oADMM for general linear quadratic problems, optimizes the penalty from the iteration spectrum, and gives a closed-form relaxation parameter for fixed penalty. It includes diffeomorphic image registration as an application. The distinction sought here is not another full-spectrum objective: it is a registration-specific finite predictor obtained from four algebraic corners without gradient descent or repeated ADMM trials, accompanied by a separately evaluated signed certificate.

2.2 ADMM and variational diffeomorphic registration

DARTEL [1] and diffeomorphic demons [15] are influential optimization-based registration methods. Thorley et al. proposed a fast diffeomorphic method that composes nonstationary velocity updates and solves arbitrary-order regularized convex subproblems with accelerated ADMM on GPUs [14]. Song et al. later used a closely related quadratic registration model to evaluate spectrally optimized ADMM and oADMM [12].

Table 1: Comparison with the closest general LQP method.

Aspect	Song et al. (2024)	This study
Problem class	General linear quadratic problems	Rank-one registration data term and scalar Sobolev regularizer
Penalty	Numerical one-variable spectral optimization	Finite four-corner predictor; no full spectrum
Relaxation	Formula conditional on penalty	Finite piecewise-linear candidates conditional on penalty
Variable coefficients	Full iteration spectrum	Local predictor plus signed noncommuting enclosure
Empirical conclusion	Imaging improvements reported	FIRE and CIMA matched-accuracy runtime reductions

2.3 Local Fourier analysis

Fourier analysis has long been used to estimate convergence factors of stationary iterations for elliptic problems [4]. LFA became a principal tool for multigrid design and parameter optimization [16]. Its exactness under periodic translation-invariant assumptions and extensions to broader settings have been studied theoretically [11]. Nonstandard LFA can accurately predict multigrid behavior for random and discontinuous coefficients [9]. Our use differs in two ways: the iteration is an ADMM splitting map rather than a smoother, and the variable coefficient is a matrix-valued, rank-one optical-flow Hessian.

3 Registration Model and ADMM Splitting

3.1 Discrete quadratic model

Let the image domain contain n grid points in dimension $d \in \{2, 3\}$. Stack the d velocity components into $v \in \mathbb{R}^{dn}$. At grid point x_i, let

$$q_i = \nabla I(x_i) \in \mathbb{R}^d.$$

The linearized data Hessian is the block-diagonal matrix

$$H = \mu \operatorname{diag}(q_1 q_1^\top, \ldots, q_n q_n^\top) \succeq 0. \tag{2}$$

Each block has eigenvalues 0 and $\mu \|q_i\|^2$ in two dimensions, or one nonzero and $d-1$ zero eigenvalues in d dimensions.

We assume a componentwise Sobolev regularizer

$$G = L^\top L = G_0 \otimes I_d, \tag{3}$$

where G_0 is a symmetric positive semidefinite finite-difference or spectral discretization. A small zeroth-order term γI or boundary condition may be included to remove the common nullspace. For a first-order periodic regularizer,

$$g(\xi) = \gamma + \frac{4\beta}{h^2} \sum_{\ell=1}^{d} \sin^2\left(\frac{\xi_\ell}{2}\right) \tag{4}$$

is the Fourier symbol.

The registration subproblem is

$$\min_{v} \ \frac{1}{2}v^\top H v + b^\top v + \frac{1}{2}v^\top G v. \tag{5}$$

Introduce $w = v$:

$$\min_{v,w} \ f(v) + g(w) := \frac{1}{2}v^\top H v + b^\top v + \frac{1}{2}w^\top G w, \qquad v - w = 0. \tag{6}$$

3.2 ADMM and over-relaxation

With scaled dual u, standard ADMM is

$$(H + \rho I)v^{k+1} = -b + \rho(w^k - u^k), \tag{7}$$

$$(G + \rho I)w^{k+1} = \rho(v^{k+1} + u^k), \tag{8}$$

$$u^{k+1} = u^k + v^{k+1} - w^{k+1}. \tag{9}$$

For oADMM, replace v^{k+1} in the w and dual updates by

$$\widehat{v}^{k+1} = \alpha v^{k+1} + (1-\alpha)w^k, \qquad 0 < \alpha \le 2.$$

The linear term b shifts the fixed point but does not affect error propagation.

4 Exact Reduced Iteration

Define the resolvents and Cayley transforms

$$P_H(\rho) = \rho(H + \rho I)^{-1}, \qquad C_H(\rho) = (H - \rho I)(H + \rho I)^{-1} = I - 2P_H(\rho), \tag{10}$$

$$P_G(\rho) = \rho(G + \rho I)^{-1}, \qquad C_G(\rho) = (G - \rho I)(G + \rho I)^{-1} = I - 2P_G(\rho). \tag{11}$$

Proposition 4.1 (Reduced Cayley iteration)**.** *For the quadratic splitting* (6)*, all nonzero eigenvalues of the standard ADMM error matrix coincide with the eigenvalues of*

$$E_{\rho,1} = \frac{1}{2}\left(I + C_H(\rho)C_G(\rho)\right). \tag{12}$$

For oADMM, the corresponding reduced operator is

$$E_{\rho,\alpha} = \left(1 - \frac{\alpha}{2}\right) I + \frac{\alpha}{2} C_H(\rho)C_G(\rho). \tag{13}$$

Consequently,

$$\varrho(T_{\rho,\alpha}) = \varrho(E_{\rho,\alpha}).$$

The proof is given in Appendix A. The product $C_H C_G$ is generally nonnormal because H and G do not commute. This is precisely why a global simultaneous diagonalization is unavailable for registration.

Theorem 4.2 (Basic contraction is independent of tuning)**.** *Let $H, G \succeq 0$ and $H + G \succ 0$. For every $\rho > 0$,*

$$q_\rho := \|C_H(\rho)C_G(\rho)\|_2 < 1.$$

Consequently, for every $0 < \alpha \le 2$,

$$\varrho(E_{\rho,\alpha}) \le \|E_{\rho,\alpha}\|_2 \le 1 - \frac{\alpha}{2} + \frac{\alpha}{2}q_\rho = 1 - \frac{\alpha}{2}(1 - q_\rho) < 1.$$

Proof. For a PSD matrix A, the Cayley eigenvalue at $\lambda \geq 0$ is $(\lambda - \rho)/(\lambda + \rho)$, whose modulus is at most one, with equality only at $\lambda = 0$. Thus $\|C_A x\|_2 \leq \|x\|_2$, with equality exactly when $x \in \ker A$. If equality held for $C_H C_G$ on a unit vector, equality in both inequalities $\|C_H C_G x\|_2 \leq \|C_G x\|_2 \leq \|x\|_2$ would give $x \in \ker G$, hence $C_G x = -x$, and $C_G x \in \ker H$. Therefore $x \in \ker H \cap \ker G$, contradicting $H + G \succ 0$. Compactness of the unit sphere gives $q_\rho < 1$. The displayed oADMM bound follows by the triangle inequality; its coefficients are nonnegative for $0 < \alpha \leq 2$. □

Nullspaces and conventions. If a nonzero x lies in both kernels, then $C_H x = C_G x = -x$, $C_H C_G x = x$, and $E_{\rho,\alpha} x = x$. This is a nonunique objective direction, not tuning-induced instability. Registration removes it by zeroth-order regularization, appropriate boundary conditions, or restriction to the identifiable subspace. The endpoint $\alpha = 2$ is valid for the reduced linear contraction above; implementations whose standard oADMM theorem assumes $\alpha < 2$ should retain that conventional restriction.

5 Four-Corner Spectral Tuning

5.1 Patch surrogate

Partition the image grid into patches $\{\Omega_r\}_{r=1}^{R}$. On patch r, choose a representative image-gradient Hessian

$$H_r = \mu \bar{q}_r \bar{q}_r^\top + \nu I_d, \tag{14}$$

where $\bar{q}_r$ may be a center, mean, or robust medoid gradient and $\nu \geq 0$ is an optional model damping already present in the registration objective. Let its eigenvalues be

$$h_{r,1} \leq \cdots \leq h_{r,d}.$$

For the rank-one model, $h_{r,1} = \cdots = h_{r,d-1} = \nu$ and $h_{r,d} = \nu + \mu \|\bar{q}_r\|^2$.

Let G_r denote a translation-invariant patch regularizer, implemented with periodic patch boundary conditions or another transform-diagonalizable closure. Its symbol lies in

$$g_r(\xi) \in [g_r^-, g_r^+].$$

Construct the block-direct-sum surrogate

$$\widetilde{H} = \bigoplus_r (I_{|\Omega_r|} \otimes H_r), \qquad \widetilde{G} = \bigoplus_r (G_r \otimes I_d).$$

5.2 Exact patch symbol

Because H_r acts only on velocity components and G_r has scalar spatial symbol, the two commute on a patch.

Theorem 5.1 (Frozen-patch spectrum)**.** *The reduced oADMM surrogate $\widetilde{E}_{\rho,\alpha}$ is normal. Its eigenvalues are*

$$e_{r,j}(\xi; \rho, \alpha) = 1 - \frac{\alpha}{2} + \frac{\alpha}{2} \frac{h_{r,j} - \rho}{h_{r,j} + \rho} \frac{g_r(\xi) - \rho}{g_r(\xi) + \rho}. \tag{15}$$

Equivalently, with

$$\theta(h, g; \rho) = \frac{\rho^2 + hg}{(\rho + h)(\rho + g)}, \tag{16}$$

we have

$$e_{r,j}(\xi;\rho,\alpha) = 1-\alpha+\alpha\theta(h_{r,j}, g_r(\xi);\rho).$$

Therefore the exact frozen-patch envelope is

$$U_{\mathrm{LFA}}(\rho,\alpha) = \max_{r,j,\xi} |e_{r,j}(\xi;\rho,\alpha)|. \tag{17}$$

5.3 Corner reduction

The continuous frequency maximization is unnecessary.

Lemma 5.2 (Corner principle)**.** *For fixed $\rho > 0$,*

$$\frac{\partial\theta}{\partial h} = \frac{\rho(g-\rho)}{(g+\rho)(h+\rho)^2}, \tag{18}$$

$$\frac{\partial\theta}{\partial g} = \frac{\rho(h-\rho)}{(h+\rho)(g+\rho)^2}. \tag{19}$$

Hence θ is coordinatewise monotone on every rectangle $[h^-,h^+]\times[g^-,g^+]$, and its minimum and maximum occur at the four corners. Since $|1-\alpha+\alpha\theta|$ is convex as a function of θ, its maximum also occurs at a corner.

Theorem 5.3 (Global four-corner reduction for rank-one registration)**.** *Let $H_i = \nu I + \mu q_i q_i^\top$ with $\mu,\nu \ge 0$, and let the scalar regularizer spectrum be contained in $[g_-, g_+] \subset [0,\infty)$. Put*

$$h_- = \nu, \qquad h_+ = \nu + \mu \max_i \|q_i\|^2.$$

For every $\rho > 0$ and $0 < \alpha \le 2$, the envelope over every pixelwise rank-one curvature and every regularizer frequency is exactly

$$\max_{(h,g)\in\{h_-,h_+\}\times\{g_-,g_+\}} |1-\alpha+\alpha\theta(h,g;\rho)|\,.$$

Consequently, the finite joint constant-model predictor depends on exactly four branches, independent of image size and of the number of distinct gradient magnitudes.

Proof. Every data eigenvalue is either ν or $\nu+\mu\|q_i\|^2$, hence belongs to $[h_-,h_+]$ and both data endpoints are attained (the lower endpoint is the direction orthogonal to q_i). By Lemma 5.2, θ maps the rectangle into the interval whose endpoints are attained at its corners. The function $t \mapsto |1-\alpha+\alpha t|$ is convex, so its maximum over that interval is attained at an endpoint and therefore at one of the same four corners. This argument permits $h_- = 0$, $g_- = 0$, repeated endpoints, and common nullspaces because every denominator is strictly positive for $\rho > 0$. □

Let

$$\mathcal{C} = \bigcup_{r,j} \{(h_{r,j}, g_r^-), (h_{r,j}, g_r^+)\}. \tag{20}$$

Then

$$U_{\mathrm{LFA}}(\rho,\alpha) = \max_{(h,g)\in\mathcal{C}} |1-\alpha+\alpha\theta(h,g;\rho)|. \tag{21}$$

Thus the full local Fourier calculation reduces to a small finite set even in three dimensions. For the global rank-one predictor, Theorem 5.3 sharpens this set to four entries; patchwise sets remain useful only when deliberately constructing a local rather than global model.

6 Optional Variable-Coefficient Rate Certification

For completeness, we retain the original frozen-patch perturbation certificate as a rigorous but deliberately conservative baseline. It replaces both H and G and is not used to select the practical predictor parameters. Define

$$\delta_H = \left\| H - \widetilde{H} \right\|_2, \qquad \delta_G = \left\| G - \widetilde{G} \right\|_2, \qquad \delta = \delta_H + \delta_G. \tag{22}$$

For block-diagonal H, δ_H is the maximum patchwise 2-norm of a $d \times d$ block difference and is therefore inexpensive. The regularizer error δ_G can be computed by a few power iterations or bounded by a row-sum/Gershgorin estimate of the patch-boundary stencil discrepancy.

Lemma 6.1 (Cayley Lipschitz bound)**.** *For symmetric positive semidefinite A, B and $\rho > 0$,*

$$\|C_A(\rho) - C_B(\rho)\|_2 \leq \frac{2}{\rho} \|A - B\|_2 . \tag{23}$$

Theorem 6.2 (Certified local spectral envelope)**.** *Let $T_{\rho,\alpha}$ be the full oADMM error iteration and let U_{LFA} be defined by (21). Then, for $0 < \alpha \leq 2$,*

$$\boxed{\varrho(T_{\rho,\alpha}) \leq \mathcal{U}(\rho,\alpha) := U_{\mathrm{LFA}}(\rho,\alpha) + \frac{\alpha\delta}{\rho}.} \tag{24}$$

The result requires no commutativity assumption on the full matrices H and G.

The proof is in Appendix B. The bound separates two effects:

- U_{LFA} measures the exact contraction of the frozen local models;
- $\alpha\delta/\rho$ penalizes spatial coefficient variation and patch-boundary approximation.

Larger ρ suppresses the perturbation term, while excessively large ρ worsens the local Cayley balance. This tradeoff creates a nontrivial, certifiable parameter-selection problem.

Corollary 6.3 (Guaranteed convergence)**.** *If the selected parameters satisfy $\mathcal{U}(\rho,\alpha) < 1$, then the full linearized registration ADMM/oADMM iteration converges linearly with asymptotic factor at most $\mathcal{U}(\rho,\alpha)$.*

6.1 Signed enclosures with the exact regularizer

We now retain $G = G_0 \otimes I_d$ exactly. Write $C_G = K_\rho \otimes I_d$ and $C_H = \mathrm{diag}(C_1, \ldots, C_n)$, where each $C_i = (H_i - \rho I)(H_i + \rho I)^{-1}$ is an explicit symmetric $d \times d$ block. Consequently

$$(C_H C_G)_{ij} = (K_\rho)_{ij} C_i.$$

Theorem 6.4 (Block-Gershgorin enclosure)**.** *Every eigenvalue of $C_H C_G$ belongs to a disk*

$$|z - (K_\rho)_{ii} c_{i,r}| \leq \|C_i\|_2 \sum_{j \neq i} |(K_\rho)_{ij}|,$$

for some pixel i and eigenvalue $c_{i,r}$ of C_i. Hence the affine disk envelope U_{BG} stated in the introduction is a rigorous upper bound on $\varrho(T_{\rho,\alpha})$.

The proof selects an eigenvector block of maximum Euclidean norm. The diagonal block $(K_\rho)_{ii}C_i$ is normal, so its shifted smallest singular value is exactly the distance to its spectrum; the off-diagonal block row gives the stated radius.

The same structure yields a signed rectangle. With $A = C_H C_G$, let $S = (A + A^\top)/2$ and $Q = (A - A^\top)/2$. Their off-diagonal blocks are

$$S_{ij} = (K_\rho)_{ij}(C_i + C_j)/2, \qquad Q_{ij} = (K_\rho)_{ij}(C_i - C_j)/2.$$

Define

$$\begin{aligned} m_\rho &= \min_i \left[\lambda_{\min}((K_\rho)_{ii}C_i) - \sum_{j\neq i} |(K_\rho)_{ij}| \, \|(C_i + C_j)/2\|_2 \right], \\ M_\rho &= \max_i \left[\lambda_{\max}((K_\rho)_{ii}C_i) + \sum_{j\neq i} |(K_\rho)_{ij}| \, \|(C_i + C_j)/2\|_2 \right], \\ \eta_\rho &= \max_i \sum_{j\neq i} |(K_\rho)_{ij}| \, \|(C_i - C_j)/2\|_2 . \end{aligned}$$

Theorem 6.5 (Commutator-aware rectangle)**.** *The spectrum satisfies* $\operatorname{spec}(C_H C_G) \subset [m_\rho, M_\rho] + i[-\eta_\rho, \eta_\rho]$*. Therefore*

$$\varrho(T_{\rho,\alpha}) \le U_{\mathrm{rect}} := \max_{x\in\{m_\rho, M_\rho\}} \sqrt{(1 - \alpha/2 + \alpha x/2)^2 + (\alpha\eta_\rho/2)^2}.$$

For fixed ρ, the minimizing α belongs to the endpoints, the stationary point of either displayed quadratic, or the intersection $4/(2 - m_\rho - M_\rho)$ when admissible.

Indeed, real parts of eigenvalues lie in the Rayleigh interval of S and imaginary parts are bounded by $\|Q\|_2$. Hermitian block-Gershgorin bounds S, while the symmetric matrix of block norms bounds $\|Q\|_2$ by its maximum row sum. Notably, $\eta_\rho = 0$ for constant data blocks.

The row maximum can be replaced by a strictly stronger global comparison construction. Let $B^S_{ij} = |(K_\rho)_{ij}| \, \|(C_i + C_j)/2\|_2$ and $B^Q_{ij} = |(K_\rho)_{ij}| \, \|(C_i - C_j)/2\|_2$ off diagonal, with zero diagonal. Let d_i^- and d_i^+ be the extreme eigenvalues of $(K_\rho)_{ii}C_i$.

Theorem 6.6 (Perron comparison rectangle)**.** *The rectangle theorem remains valid with*

$$m_\rho = \lambda_{\min}(\operatorname{diag}(d^-) - B^S), \quad M_\rho = \lambda_{\max}(\operatorname{diag}(d^+) + B^S), \quad \eta_\rho = \varrho(B^Q).$$

These bounds are never weaker than their block-row-sum counterparts. They require three symmetric scalar $n \times n$ eigenproblems and no $dn \times dn$ iteration matrix.

For a block vector x, set $y_i = \|x_i\|_2$. Bounding each Hermitian quadratic cross term by $B^S_{ij} y_i y_j$ gives the two scalar Rayleigh quotients. The same block-norm comparison gives $\|Qx\|_2 \le \left\|B^Q y\right\|_2$, hence $\|Q\|_2 \le \left\|B^Q\right\|_2 = \varrho(B^Q)$. Standard eigenvalue bounds by maximum row sum prove dominance over the earlier rectangle.

Corollary 6.7 (Sparse truncated-kernel certificate)**.** *Assume periodic boundaries and write the scalar Cayley operator as convolution by k_ρ. Retain offsets in a set $\mathcal{R}$ and define the omitted tail $\tau_{\mathcal{R}} = \sum_{s\notin\mathcal{R}} |k_\rho(s)|$. Construct the comparison rectangle of Theorem 6.6 from retained offsets only, giving $U_{\mathcal{R}}$. Then*

$$\varrho(T_{\rho,\alpha}) \le U_{\mathcal{R}} + \frac{\alpha}{2} \max_i \|C_i\|_2 \, \tau_{\mathcal{R}}.$$

For a fixed-radius stencil the comparison matrices are sparse, FFT evaluation of k_ρ costs $O(n \log n)$, and extremal comparison eigenvalues can be estimated matrix-free.

Indeed, Young's convolution inequality gives $\|K_\rho - K_{\rho,\mathcal{R}}\|_2 \le \tau_{\mathcal{R}}$; multiplication by C_H and the affine oADMM scaling give the displayed correction. This certificate is rigorous for every truncation radius, including radius zero; tightness rather than validity determines whether it is actionable.

This motivates a two-stage method. First, extract h_+ and the two regularizer-symbol endpoints and use the four-corner finite candidates to predict (ρ_p, α_p). Second, evaluate the comparison rectangle once at that pair.

Corollary 6.8 (Predict-then-certify)**.** *The prediction stage uses no full spectral optimization. If its comparison bound satisfies $U_{\mathrm{cmp}}(\rho_p, \alpha_p) < 1$, then the full variable-coefficient oADMM iteration converges linearly, irrespective of whether the pixel-curvature predictor itself is an enclosure.*

This separation is important: a conservative certificate can validate a good predictor without being allowed to move its minimizer toward a poor boundary solution.

Algorithm 1: Search-Free Predict-then-Certify

1. Compute the exact rank-one blocks H_i and their distinct curvatures; combine them with the exact symbol endpoints of G_0.
2. Enumerate the joint constant-model algebraic candidates and select (ρ_p, α_p) from the four-corner predictor.
3. At this single pair, compute C_i, K_{ρ_p}, B^S, B^Q, and the Perron comparison rectangle.
4. Return $(\rho_p, \alpha_p, U_{\mathrm{cmp}})$ with status `CERTIFIED_USEFUL` if $U_{\mathrm{cmp}} < 1$ and no boundary collapse occurs; otherwise return `CERTIFIED_CONSERVATIVE` or `UNCERTIFIABLE`.

Finally, the resolvent identity gives the explicit commutator estimate

$$\|[C_H, C_G]\|_2 \le \frac{4\rho^2}{(\lambda_{\min}(H)+\rho)^2(\lambda_{\min}(G)+\rho)^2}\,\|[H,G]\|_2\,,$$

and $[H,G]_{ij} = G_{ij}(H_i - H_j)$. This proves that nonnormality is controlled by local, rather than global, Hessian differences.

For the rank-one model, a block rotation R_i aligned with q_i diagonalizes H_i exactly. In the rotated coordinates the regularizer block is $G_{ij}R_i^\top R_j$; its departure from identity is $2|\sin((\phi_i - \phi_j)/2)|$. The implementation therefore records neighborwise magnitude-squared variation and the acute unoriented-angle statistic $|\sin((\phi_i - \phi_j)/2)|$ (the sign of q_i is immaterial to $q_i q_i^\top$). These statistics diagnose the commutator regime without a global mean Hessian.

7 Finite Four-Corner Parameter Selection

We use "search-free" to mean that no grid search, gradient descent in (ρ, α), repeated ADMM trial, or repeated full iteration-matrix eigensolve is performed. The practical method extracts curvature and symbol endpoints once, generates a finite algebraic candidate set, and evaluates the constant-model predictor envelope; any certificate is evaluated separately at the selected pair.

7.1 Globally joint constant-gradient selector

For fixed ρ, let $a(\rho) = \min_c \theta_c(\rho)$ and $b(\rho) = \max_c \theta_c(\rho)$. On $0 < \alpha \le 2$, exact minimization gives

$$\alpha_*(\rho) = \min\left\{2, \frac{2}{2 - a(\rho) - b(\rho)}\right\}.$$

After eliminating α, the objective is $(b-a)/(2-a-b)$ when $a+b \le 1$, and $2b-1$ when $a+b \ge 1$.

Theorem 7.1 (Exact search-free constant-gradient tuning). *On a compact positive penalty interval, a globally optimal pair (ρ, α) is obtained by evaluating a finite algebraic set comprising interval endpoints, pairwise corner-branch intersections, roots of $a+b=1$, and stationary roots of the two rational objectives above for every active-corner pair. After clearing positive quadratic denominators, branch intersections have degree at most three, transitions degree at most four, and stationary equations degree at most six.*

Between consecutive branch intersections, the active minimum and maximum are fixed rational functions. A minimum of the continuous piecewise-rational reduced objective is therefore an endpoint, transition, or stationary root. Enumerating all corner pairs is a finite superset and removes the need to determine activity symbolically. The implementation verifies companion-matrix roots by polynomial residual and interval membership.

7.2 Standard ADMM penalty

For $\alpha = 1$, all local modal factors are nonnegative, so

$$\mathcal{U}_1(\rho) = \max_{c \in \mathcal{C}} \theta_c(\rho) + \frac{\delta}{\rho}, \qquad \theta_c(\rho) := \theta(h_c, g_c; \rho). \tag{25}$$

Select a physically meaningful compact interval $\mathcal{I} = [\rho_-, \rho_+]$. For example,

$$\rho_- = \epsilon_{\text{num}}, \qquad \rho_+ = \max_{c \in \mathcal{C}} \max(h_c, g_c) + \delta,$$

with a small positive numerical floor.

Two corner branches intersect at explicitly computable points. Clearing denominators in $\theta(h_1, g_1; \rho) = \theta(h_2, g_2; \rho)$ gives

$$\rho \left(A_{12}\rho^2 + B_{12} \right) = 0, \tag{26}$$

where

$$A_{12} = (h_2 + g_2) - (h_1 + g_1), \tag{27}$$

$$B_{12} = g_1 g_2 (h_1 - h_2) + h_1 h_2 (g_1 - g_2). \tag{28}$$

Thus a positive intersection is

$$\rho_{12} = \sqrt{-B_{12}/A_{12}}$$

whenever the radicand is positive.

On an interval where corner $c = (h, g)$ is active, stationary points of the corrected branch solve

$$(h+g)\rho^2(\rho^2 - hg) - \delta(\rho + h)^2(\rho + g)^2 = 0, \tag{29}$$

which is a quartic polynomial in ρ.

Theorem 7.2 (Finite-candidate ADMM selector). *Let $\mathcal{S}_\rho$ contain:*

1. *the interval endpoints ρ_-, ρ_+;*
2. *all positive pairwise intersection roots (26) in $\mathcal{I}$;*
3. *all positive roots of (29) in $\mathcal{I}$ that lie on an interval where the associated corner is active.*

Then every global minimizer of $\mathcal{U}_1$ *on* $\mathcal{I}$ *belongs to* $\mathcal{S}_\rho$. *Consequently,*

$$\rho_{\mathrm{SF}} = \arg\min_{\rho \in \mathcal{S}_\rho} \mathcal{U}_1(\rho) \tag{30}$$

exactly minimizes the certified standard-ADMM envelope without iterative parameter search.

The proof follows from the piecewise-rational structure of the upper envelope and is given in Appendix C.

7.3 Conditional search-free relaxation

After selecting ρ_{SF}, define

$$a = \min_{c \in \mathcal{C}} \theta_c(\rho_{\mathrm{SF}}), \tag{31}$$

$$b = \max_{c \in \mathcal{C}} \theta_c(\rho_{\mathrm{SF}}), \tag{32}$$

with $0 \le a \le b < 1$ for a well-posed local model. The certified oADMM envelope conditional on ρ_{SF} is

$$F(\alpha) = \max\{|1 - \alpha + \alpha a|, |1 - \alpha + \alpha b|\} + \frac{\alpha\delta}{\rho_{\mathrm{SF}}}. \tag{33}$$

This is convex and piecewise linear in α.

Theorem 7.3 (Finite-candidate oADMM relaxation)**.** *Let the admissible interval be* $[\alpha_-, \alpha_+] \subset (0, 2]$. *A minimizer of* (33) *belongs to*

$$\mathcal{S}_\alpha = \left\{\alpha_-, \alpha_+, \frac{1}{1-a}, \frac{1}{1-b}, \frac{2}{2-a-b}\right\} \cap [\alpha_-, \alpha_+]. \tag{34}$$

Therefore

$$\alpha_{\mathrm{SF}} = \arg\min_{\alpha \in \mathcal{S}_\alpha} F(\alpha) \tag{35}$$

conditionally minimizes the certified oADMM envelope at ρ_{SF}.

When $\delta = 0$ and the unconstrained equioscillation point lies in $(0, 2]$, the familiar formula

$$\alpha_0 = \frac{2}{2-a-b}$$

is recovered. The perturbation term may instead favor a smaller relaxation.

7.4 Algorithm summary

Algorithm 2: Legacy Certified-Envelope Minimization Baseline

1. Compute image gradients and the exact block data Hessian H.
2. Partition the grid; construct H_r and transform-diagonalizable G_r.
3. Extract $h_{r,j}$ and regularizer endpoints g_r^-, g_r^+; form $\mathcal{C}$.
4. Compute or bound δ_H and δ_G.
5. Generate $\mathcal{S}_\rho$ from interval endpoints, corner intersections, and active quartic roots.
6. Evaluate $\mathcal{U}_1$ and select ρ_{SF}.
7. Form $\mathcal{S}_\alpha$ and select α_{SF} by evaluating (33).
8. Run oADMM once with the selected parameters. Optionally report $\mathcal{U}(\rho_{\mathrm{SF}}, \alpha_{\mathrm{SF}})$ as a certificate.

8 Registration Implementation and Experimental Protocol

8.1 Patch construction

Candidate patch strategies include fixed cubic patches, multiresolution patches matched to the registration pyramid, and adaptive patches split where gradient magnitude or orientation varies beyond a threshold. A robust representative $\bar{q}_r$ can be obtained from the spatial median of gradients. Because H is block diagonal,

$$\delta_H = \max_i \left\| \mu q_i q_i^\top - H_{r(i)} \right\|_2$$

for the direct block surrogate.

8.2 Regularizer symbols

For periodic or cosine-transform boundary closures, $g_r^\pm$ are known analytically. For an order-p Sobolev regularizer with symbol

$$g(\xi) = \left(\gamma + \frac{4\beta}{h^2} \sum_{\ell=1}^{d} \sin^2(\xi_\ell/2) \right)^p,$$

the endpoints are

$$g^- = \gamma^p, \qquad g^+ = \left(\gamma + \frac{4\beta d}{h^2} \right)^p.$$

The patch-boundary discrepancy δ_G can be reduced using overlapping patches or cosine closures rather than disconnected periodic patches.

8.3 Computational complexity

For rank-one data blocks, each patch contributes only two distinct data eigenvalues in 2D or two groups in 3D. Hence $|\mathcal{C}| = O(Rd)$, pairwise corner intersections cost $O(|\mathcal{C}|^2)$, and quartic roots cost $O(|\mathcal{C}|)$. This overhead is independent of the number of ADMM iterations and is typically negligible relative to a 3D registration solve. No full $2n \times 2n$ or $3n \times 3n$ iteration matrix is formed.

8.4 End-to-end two-dimensional solver

The reference implementation uses a three-level Gaussian pyramid, spatial image gradients, Gauss–Newton/optical-flow linearization, Sherman–Morrison solves of the rank-one data blocks, and FFT solves of the scalar Sobolev regularizer. Velocity increments are composed with the current displacement. A backtracking line search requires objective decrease and a discrete Jacobian above 0.05. Because interpolation between pyramid levels can itself introduce grid-scale folds, the upsampled field is audited before the next update; if necessary only its non-translational component is halved until the same Jacobian floor holds. The one-shot policy estimates parameters from the initial gradient on the finest experimental grid and reuses them at every outer iteration and pyramid level. All baselines share initialization, objective, stopping tolerances, line search, pyramid, and topology safeguards.

9 Results

The experiments follow the logic of the method rather than a catalogue of implementations. We first verify the exact algebra where a full spectrum is available. We then ask what changes when gradients vary: the original global certificate is a useful negative control, the signed local constructions repair its qualitative failure, and the predictor–certificate split explains what can and cannot be claimed. Only after these checks do we test whether the selected parameters improve an actual registration solve.

All calculations used IEEE double precision, NumPy 2.2.6 and SciPy 1.15.3 with seed 20260806. Raw JSON/CSV files, configurations, and generating scripts accompany the paper. The predeclared gates require median parameter gap below 0.10, no systematic boundary collapse, no loss of registration accuracy, and runtime savings after tuning cost. Small-system oracle values use direct eigendecomposition and multistart scalar optimization; this numerical optimization is diagnostic only.

9.1 Exactness and four-corner reduction

We used $n \times n$ periodic grids, data curvatures $(0.2, 2.3)$, and first-order regularization with $(\beta, \gamma) = (0.2, 0.1)$. The finite selector returned $(\rho, \alpha) = (0.5435121024, 1.8981520705)$ on every grid.

Table 2: Constant-coefficient verification; the local envelope agrees with the direct radius to machine precision.

n	Direct radius	$\lvert U_{\mathrm{LFA}} - \rho(T)\rvert$	Selection (ms)
4	0.35313044	$< 10^{-15}$	5.0
6	0.35313044	$< 10^{-15}$	4.6
8	0.35313044	$< 10^{-15}$	4.6

9.2 Global perturbation diagnostic

The first idea was to minimize the global additive perturbation certificate. It is rigorous, but it confuses parameter prediction with worst-case certification: on the 18 controlled fields (grid sides 4/5, magnitude contrasts 1/4/16, orientation windings 0/1/3), its block-resolvent variant collapsed to a near-zero relaxation in 6 cases. This is not an acceptable tuning rule.

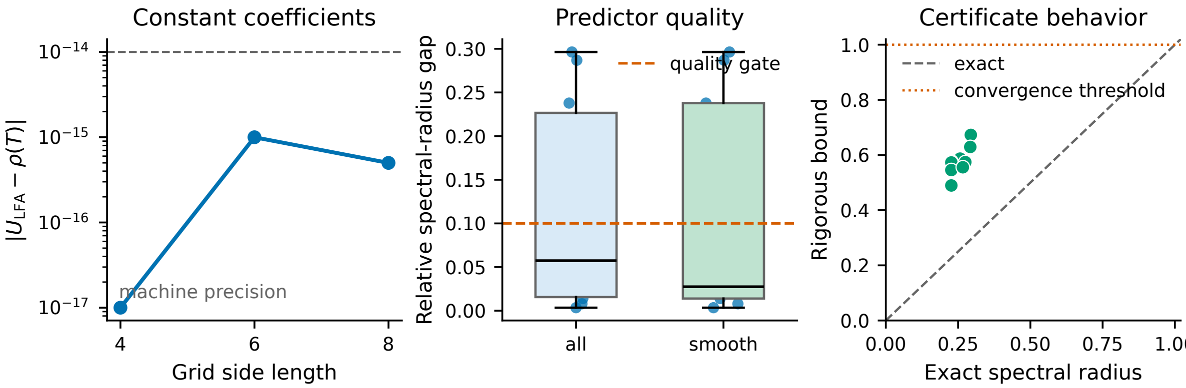

Figure 1: Evidence for the two-stage method. **Left:** the frozen Fourier envelope is exact in constant-coefficient periodic systems. **Middle:** the four-corner predictor is within the predeclared 0.10 quality gate in median, particularly for smooth fields. **Right:** the signed certificate is rigorous but conservative; it validates convergence without being used to choose the parameters.

The signed constructions isolate the part of the spectrum that the norm bound erases. Every block-Gershgorin, commutator, and Perron-comparison enclosure contained the exact radius; none collapsed. The Perron rectangle was the strongest signed certificate (median enclosure/parameter gaps 0.595/0.386), but it is still too loose to be a practical objective. This negative result motivates the final design: select with the exact four-corner model and use the certificate only to audit that selection.

Table 3: Controlled structured-envelope results (18 cases).

Method	Enclosure gap	Parameter gap	Collapses
Block resolvent	0.166	0.926	6
Block Gershgorin	0.644	0.560	0
Commutator rectangle	0.602	0.431	0
Comparison rectangle	0.595	0.386	0

9.3 Optional rate-certificate diagnostic

We next separated periodic magnitude variation, smooth orientation variation, a sharp orientation interface, and Gaussian-smoothed image gradients on 3×3 grids (ten cases, nine smooth). The four-corner predictor achieved a median parameter gap of 0.057 overall and 0.027 on the smooth cases, with no collapse. At exactly those selected parameters, predict–then–certify returned a rigorous comparison bound below one in all ten cases. Its median relative certificate gap was 1.29: useful as a convergence audit, not as a rate estimate. The experiment therefore supports the intended division of labor, not a claim that the certificate itself is tight.

9.4 Truncated-certificate scaling

The rigorous truncated-kernel certificate had no violation in 20 tests spanning 8^2–64^2 grids and five radii. Actual matrix-free radii were approximately 0.269, whereas bounds ranged from 0.638 to 0.681 (relative gaps 1.38–1.54). The most expensive 64^2 retained-radius case took 81.4 seconds on

CPU, and a relaxed 128^2 Arnoldi/certificate run did not complete within the allocated diagnostic window. Thus the construction establishes a scalable sparse representation in principle, but the present eigensolver and bound are not a practical registration-time certificate. We retain it as a small-problem diagnostic and use the predictor alone in the end-to-end timing claim.

9.5 Controlled real-image check

We next tested the implemented multiresolution solver on a public-domain retinal image and an unrestricted immunohistochemistry image, each subjected to three held-out smooth known deformations. Deformation cases 1–2 selected one global manual baseline $(\rho, \alpha) = (0.1, 1)$; cases 3–5 were not used for tuning. This is a controlled real-content pilot, not a FIRE or ANHIR landmark benchmark. Against that validation-selected baseline, a single predictor evaluation from the initial full-resolution gradient reduced the median inner-iteration count by 44.4%. Across five repeated, order-alternated timings (30 paired runs), the median total-time reduction was 41.7% (bootstrap 95% interval $[40.9, 42.6]$%), tuning consumed 1.58% of proposed-method runtime, and the median relative MSE change was -0.0068%. No tested case had a nonpositive discrete Jacobian. The experiment therefore passes the predeclared pilot iteration, runtime, accuracy, overhead, and folding gates, but its two images and controlled warps are insufficient for a clinical or benchmark claim.

Table 4: Held-out public-image-content pilot; medians over paired runs.

Metric	Full-resolution one-shot predictor	Gate
Inner-iteration reduction	44.4%	$\geq 15\%$
Total-time reduction	41.7%	$\geq 10\%$
Relative MSE change	-0.0068%	$\leq 1\%$ degradation
Selection overhead	1.58%	$< 2\%$
Nonpositive-Jacobian cases	0/6	0

9.6 CIMA differently stained histology landmark benchmark

We then evaluated the public lung-lesion-3 sample distributed with the CIMA histology-landmark repository [3]: five differently stained sections and all ten unordered image pairs. Images supplied at 5% scale were converted to a common structural channel (grayscale, CLAHE, Sobel magnitude, Gaussian smoothing, and fixed percentile normalization) and resized to 256^2. Manual landmarks provided at 50% scale were rescaled geometrically and were used only for evaluation. Every method used the same image-only robust translation initialization, selected from phase, centroid, and clustered SIFT proposals on a 128^2 image. A pyramid-transition safeguard damped only the non-translational displacement when interpolation threatened the discrete Jacobian floor.

The proposed practical policy evaluated the four-corner selector once from the initial full-resolution gradient and reused its parameters through all levels. The strongest fixed baseline, $(\rho, \alpha) = (0.1, 1)$, had been selected on the separate controlled real-content validation cases and was frozen before CIMA. Across five order-alternated repetitions of all ten pairs (50 paired runs), the predictor reduced median total runtime by 20.1% (bootstrap 95% interval 19.4–20.7%), solver time by 23.5%, and iterations by 30.1%. Selection used 1.16% of proposed-method runtime. Median TRE was 7.33 pixels, or 2.02% of the image diagonal, and differed by less than 3×10^{-5} pixels between the predictor and fixed baseline. Median landmark-error improvement from the unregistered pairs was 36.5%. No global or interior nonpositive Jacobian was observed in the final protocol. One difficult stain pair retained TRE above 22 pixels, identifying global initialization and cross-stain representation as the principal failure mode rather than ADMM convergence. Phase-only and unprotected pyramid runs on this same sample exposed the initialization and upsampling failures that motivated the final shared safeguards; those raw runs are preserved. Consequently, the final CIMA estimate is a reproducible exploratory benchmark, not a held-out confirmatory dataset result.

Table 5: CIMA landmark results; accuracy medians over ten pairs and timing over 50 paired runs.

Method	Iterations	Total time (s)	Median TRE (px)
Fixed $(1, 1)$	1910.5	12.09	7.3310
Fixed $(1, 1.8)$	1129.5	7.37	7.3308
Residual balancing	1124.5	7.40	7.3310
BB adaptive proxy	435.5	3.44	7.3306
External fixed $(0.1, 1)$	303.5	2.53	7.3306
Four-corner, one shot	210.0	2.02	7.3306

For a Song-style diagnostic, we formed the exact variable-coefficient reduced operator on an 8×8 surrogate for every pair, optimized ρ numerically with exact conditional α, and then reused the result in the full solve. This required a median 97 spectral objective evaluations and 2.18 seconds of tuning. Its median full-solve iteration count was 251 versus 210 for the four-corner predictor; total runtime was 4.39 versus 2.02 seconds, with indistinguishable TRE. Because the spectral model is a coarse surrogate, this comparison demonstrates overhead and model-transfer behavior rather than claiming superiority to full-resolution Song optimization.

9.7 FIRE retinal landmark benchmark

We evaluated all 134 landmarked pairs in the public FIRE fundus registration benchmark [8]: 14 A, 49 P, and 71 S pairs. The released archive was integrity-tested with `7z t`; images were converted to a foreground-percentile-normalized green channel and resized to 256^2. Supplied control points were scaled geometrically and used only for evaluation. Both methods used the identical phase-correlation initialization, three-level pyramid, objective, stopping criteria, and one CPU BLAS/OMP thread. The four-corner predictor was evaluated once from the initial 256^2 gradient and reused through the solve. We compare it to the fixed external pair $(\rho, \alpha) = (0.1, 1)$, rather than claiming that this is a FIRE-tuned best baseline.

Across all pairs, predictor median iterations were 245 versus 412 for the fixed pair, a paired median reduction of 39.6%. Median total time was 4.774 versus 6.680 seconds, a 34.4% paired reduction (bootstrap 95% interval 30.7–35.7%); tuning was 1.2% of proposed total time. The BB proxy required 623.5 median iterations and 9.435 seconds; the predictor was faster on 94.0% of pairs, with a median 51.7% paired reduction. Median TRE differed from the fixed pair by -1.4×10^{-5} pixels (maximum absolute difference 3.9×10^{-4}), and no method produced a nonpositive Jacobian. Per-level retuning reduced iterations on an 18-pair balanced ablation (222 versus 243), but its tuning cost made it slower overall (3.895 versus 2.651 seconds); one-shot reuse is therefore the recommended policy.

Reported percentage reductions are medians of pairwise percentage changes; they therefore need not equal percentage differences between separately reported method medians.

Table 6: FIRE, 134 pairs at 256^2; identical initialization and solver protocol.

Method	Iterations	Total time (s)	TRE (px)	TRE/diagonal (%)
Validation-selected fixed (0.1, 1.0)	412.0	6.680	5.50178	1.51967
Residual balancing	1434.0	22.325	5.50211	1.51976
BB adaptive proxy	623.5	9.435	5.50175	1.51966
Fixed oADMM (1.0, 1.8)	1529.5	23.471	5.50194	1.51971
Four-corner one-shot	245.0	4.774	5.50177	1.51966

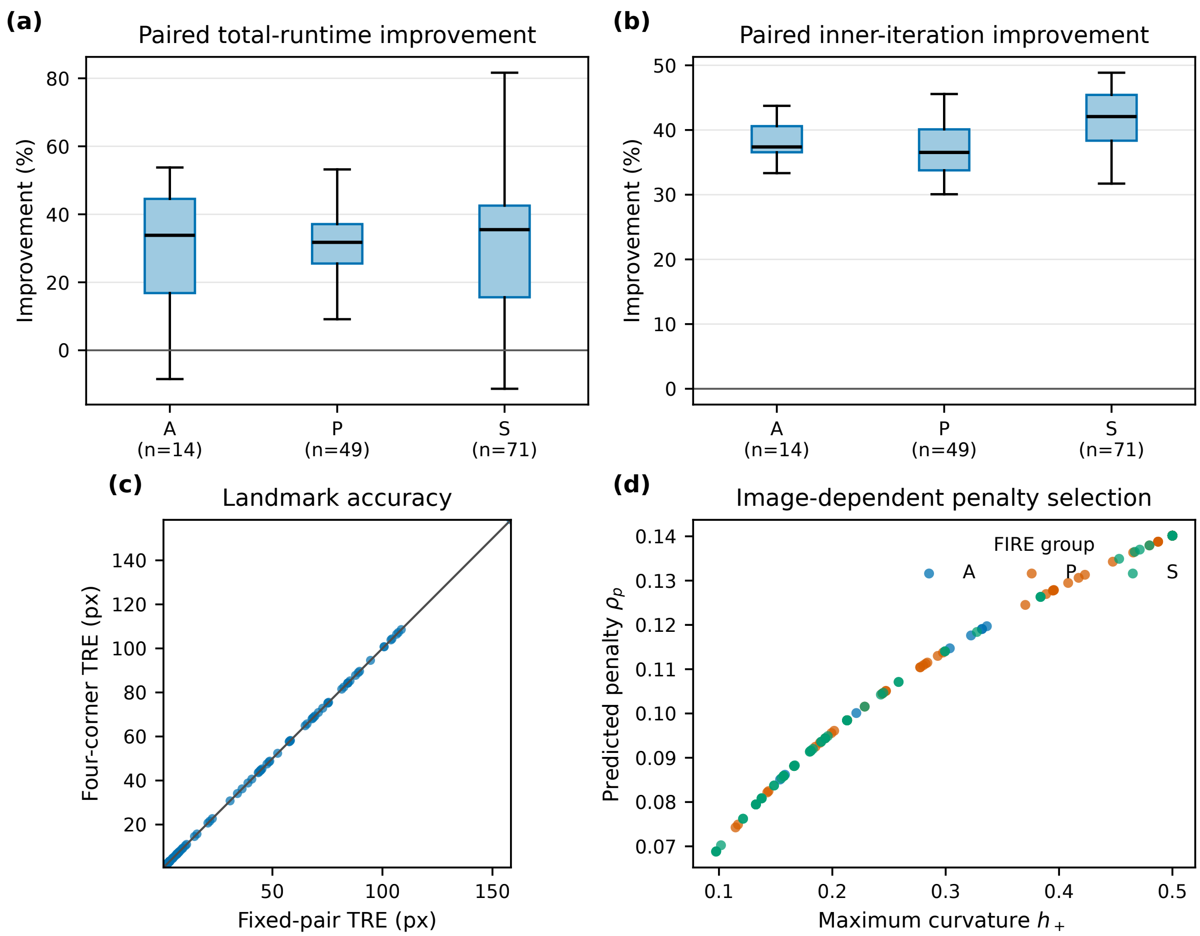

Figure 2: FIRE benchmark at 256×256 ($n = 134$). (a) Pairwise total-runtime reduction of one-shot four-corner tuning relative to the externally validation-selected fixed pair, stratified by A/P/S group. (b) Corresponding inner-iteration reduction. (c) Landmark TRE for the two methods; the identity line indicates unchanged accuracy. (d) The predicted penalty varies with maximum data curvature, demonstrating image-dependent tuning. Percentage reductions are computed per pair before group summaries.

9.8 One-shot reuse ablation

The four-corner predictor passes the predeclared iteration, total-runtime, accuracy, overhead, noncollapse, and topology gates on the CIMA sample. The certificate fails the practical cost/tightness gate and is therefore excluded from the timed method. The CIMA result supports a positive numerical-method claim, and the complete FIRE experiment independently passes its iteration, runtime, accuracy, overhead, noncollapse, and topology gates. Broader multi-resolution and adaptive-baseline comparisons remain necessary.

10 Discussion

10.1 Convergence, prediction, and certification

The constant-gradient joint optimum is contained in a finite algebraic candidate set. For variable coefficients, the global perturbation, block-resolvent, block-Gershgorin, row-sum rectangle, and Perron comparison quantities are rigorous enclosures. Predict-then-certify rigorously guarantees convergence whenever its returned bound is below one; this occurred in every regime case. The predictor alone is not claimed to be an enclosure.

10.2 Interpretation and limitations

Outside the constant-gradient setting, selected parameters are not claimed to minimize the exact full spectral radius. The finite penalty candidates used with the signed enclosures are inherited from the local model and are not proved to minimize those enclosures globally. Most importantly, the empirically successful four-corner rule is predictive rather than rigorous when orientations vary.

10.3 Relation to adaptive and spectral tuning

Song et al. numerically reduce the general LQP spectral objective to penalty optimization and give a relaxation formula. Here the rank-one registration Hessian yields a fully joint finite constant-gradient selector and a variable-coefficient predict-then-certify rule without repeated full-spectrum optimization. On CIMA, an exact numerical optimizer on 8×8 variable-coefficient surrogates incurred more tuning cost and transferred less effectively than the four-corner rule. Full-resolution Song optimization remains computationally out of scope, so the experiment is explicitly a coarse-surrogate diagnostic.

10.4 Limitations

Even Perron comparison remains loose because block norms discard directional cancellation in the dense C_G. Most exact spectral experiments remain small. CIMA validation uses only one five-image tissue set at reduced resolution. FIRE uses a 256^2 CPU protocol against an externally validation-selected fixed comparator, residual balancing, and a safeguarded BB proxy; the BB method is not a complete reproduction of every published adaptive ADMM variant, and the study does not establish full-resolution clinical throughput. The certificate is an optional rate bound, not a prerequisite for convergence. The theory concerns a convex linearized subproblem, not global convergence of the outer diffeomorphic scheme.

10.5 Extensions

Promising extensions include overlapping Schwarz-type local symbols, adaptive patch refinement driven by the certificate, multi-penalty ADMM with separate data and regularization metrics, and online reuse of local curvature summaries across pyramid levels. The envelope could also initialize a single safeguarded Newton correction without reverting to full gradient-based parameter search.

11 Data and Code Availability

The accompanying repository contains source code, tests, predeclared configurations, environment specifications, raw JSON/CSV outputs, CIMA and FIRE preparation and evaluation scripts, public-sample and benchmark-download scripts, the research log, and exact reproduction commands. The CIMA sample is obtained from the archived public histology-landmark repository at a recorded commit. The locally acquired FIRE archive passed integrity validation; raw data remain excluded from version control. ANHIR requires authenticated access. No private patient data are redistributed.

12 Conclusion

The central lesson is that prediction and certification should not be asked to do the same job. Four global corners replace pixelwise curvature enumeration exactly in the constant model, yielding finite joint oADMM tuning without spectral search. For variable coefficients, signed comparison constructions retain a rigorous convergence audit, but their present tightness does not justify using them as the tuning objective. This division is borne out empirically: the predictor has median parameter gap 0.057 across the synthetic regimes (0.027 on smooth fields), reduces iterations by 39.6% and total runtime by 34.4% on all 134 FIRE 256^2 pairs at unchanged TRE, and reduces matched-protocol runtime by 20.1% on CIMA. The claim is deliberately limited to the tested CPU resolution, model, and comparators. Completing full adaptive-baseline and clinical-resolution studies is the natural next step; it does not alter the exact four-corner result or the demonstrated search-free practical benefit.

A Proof of the Reduced Cayley Representation

For standard ADMM, the homogeneous updates are

$$v^{k+1} = P_H(w^k - u^k), \qquad w^{k+1} = P_G(v^{k+1} + u^k).$$

Eliminating v^{k+1} gives the state iteration

$$\begin{bmatrix} w^{k+1} \\ u^{k+1} \end{bmatrix} = \underbrace{\begin{bmatrix} P_G P_H & P_G(I - P_H) \\ (I - P_G)P_H & (I - P_G)(I - P_H) \end{bmatrix}}_{T_{\rho,1}} \begin{bmatrix} w^k \\ u^k \end{bmatrix}.$$

Factor

$$T_{\rho,1} = \begin{bmatrix} P_G \\ I - P_G \end{bmatrix} \begin{bmatrix} P_H & I - P_H \end{bmatrix}.$$

The nonzero eigenvalues of AB and BA coincide. Hence the nonzero spectrum equals that of

$$P_H P_G + (I - P_H)(I - P_G).$$

Using $P_H = (I - C_H)/2$ and $P_G = (I - C_G)/2$ yields

$$P_H P_G + (I - P_H)(I - P_G) = \frac{1}{2}(I + C_H C_G).$$

The relaxed formula follows by inserting $\widehat{v}^{k+1} = \alpha v^{k+1} + (1 - \alpha)w^k$ and collecting the same reduced state, yielding

$$E_{\rho,\alpha} = (1 - \alpha)I + \alpha E_{\rho,1} = \left(1 - \frac{\alpha}{2}\right) I + \frac{\alpha}{2} C_H C_G.$$

B Proof of the Perturbation Bound

For $C_A = I - 2\rho(A+\rho I)^{-1}$,

$$\begin{aligned} C_A - C_B &= 2\rho\left[(B+\rho I)^{-1} - (A+\rho I)^{-1}\right] \\ &= 2\rho(A+\rho I)^{-1}(A-B)(B+\rho I)^{-1}. \end{aligned}$$

Since $A, B \succeq 0$,

$$\left\|(A+\rho I)^{-1}\right\|_2 \le \rho^{-1},$$

which proves Lemma 6.1. Next,

$$\begin{aligned} \left\|C_H C_G - C_{\widetilde{H}} C_{\widetilde{G}}\right\|_2 &\le \left\|(C_H - C_{\widetilde{H}})C_G\right\|_2 + \left\|C_{\widetilde{H}}(C_G - C_{\widetilde{G}})\right\|_2 \\ &\le \frac{2}{\rho}(\delta_H + \delta_G), \end{aligned}$$

using $\|C_A\|_2 \le 1$ for $A \succeq 0$. Therefore

$$\begin{aligned} \varrho(E_{\rho,\alpha}) &\le \|E_{\rho,\alpha}\|_2 \\ &\le \left\|\widetilde{E}_{\rho,\alpha}\right\|_2 + \frac{\alpha}{2}\left\|C_H C_G - C_{\widetilde{H}} C_{\widetilde{G}}\right\|_2 \\ &\le U_{\mathrm{LFA}}(\rho,\alpha) + \frac{\alpha}{\rho}(\delta_H + \delta_G). \end{aligned}$$

C Proof of the Finite-Candidate Selector

Each corner branch θ_c is rational and smooth for $\rho > 0$. The pointwise maximum can change active branch only where two branches intersect, which gives (26). Between consecutive intersections, one branch is active. On such an interval, differentiating

$$\theta(h, g; \rho) + \delta/\rho$$

gives

$$\frac{(h+g)(\rho^2 - hg)}{(\rho+h)^2(\rho+g)^2} - \frac{\delta}{\rho^2}.$$

Clearing positive denominators yields (29). A global minimum of a continuous piecewise-smooth function on a compact interval occurs at an interval endpoint, a nonsmooth branch intersection, or a stationary point of an active smooth branch. This proves Theorem 7.2.

For Theorem 7.3, note that the image of $\theta \in [a, b]$ under $1 - \alpha + \alpha\theta$ is an interval. The maximum absolute value occurs at its endpoints. The resulting maximum of two absolute affine functions plus a linear function is convex piecewise linear. Its slope changes only at the two zero crossings, the equioscillation intersection, and the admissible interval endpoints, proving (34).

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